

CIS 665 – E-Business Application Technologies

**Department of Computer Information Systems
College of Business
Colorado State University**

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Instructor: Leo Vijayasathy, Ph.D.
Title: Professor
Office: 155 Rockwell Hall
Phone: 491-0607
E-Mail: Leo.Vijayasathy@colostate.edu*
Office Hours: TR – 9:30 to 10:30 AM; 3:30 to 4:30 PM

Class Days & Time: TR 11:00 – 12:15 PM
Class Location: 118 Rockwell West
Section IDs: 001 (11152), 801 (31981)

Instructional Coordinators: Hussein Al Rousan, Omar Kahwaji, Laurel Milliken
Emails: cis665sc@gmail.com, Laurel.Milliken@colostate.edu

* Please use Canvas' Messaging system for all course related communication.

Course Description

Web applications development is a complex and rapidly evolving field. There are a myriad of languages, techniques, technologies and tools that can be used to design, build and deliver a wide variety of content, services and products on the web. This course will cover some essential and popular languages/technologies for web development, namely, HTML, CSS, PHP, and ASP .Net.

Course Objective

Upon successful completion of the course, you should have the knowledge and skills to design and build functional web applications that are easy to use, interactive and dynamic.

Textbooks

I recommend that you have reference books or tutorials for HTML, PHP and ASP .Net. The bookstore should have the optional books listed below. You may choose from this set or buy others that are more suited to your learning style.

- [HTML5 and CSS3 – Comprehensive](#), Carey
- [Learning PHP, MySQL & JavaScript](#), Nixon

Software

You will need a text editor for writing your programs. TextPad, NetBeans, and Microsoft Visual Studio 2017 are available on E-Lab.

If you wish to have a copy of the IDE, you can get Visual Studio Community at <http://www.visualstudio.com> or <https://imagine.microsoft.com/en-US>.

Course Materials

All course-related materials will be available through Canvas (<http://info.canvas.colostate.edu/login.aspx>).

Course Requirements for Grades

Requirements	Points
HTML and CSS Assignment (individual)	30
Hands-on exercises (individual)	90
Project (team)	
• Proposal	20
• PHP	100
• ASP .Net	100
Research Presentation (team)	30
Midterm and Final Exams (Individual)	130
Total Points	500

The final course grade will be assigned based on the following scale:

Total Points	Letter Grade
467 and above	A
450 - 466	A-
433 - 449	B+
417 - 432	B
400 - 416	B-
383 - 399	C+
350 - 382	C
300 - 349	D
Below 300	F

Assignment

There will be an individual assignment to test your HTML and CSS knowledge and skills. Instructions for this assignment will be distributed separately.

Hands-on Exercises

I will assign a number of hands-on-exercises throughout the semester. You will complete some of these exercises in-class, and others as homework. You will have a week to complete each exercise (i.e., exercises that are assigned during a particular week, will be due the following Tuesday or Thursday at 11:59 PM). There is no makeup or extension for a missed exercise.

Project

A major component of your grades is a multi-part project that is intended to help you learn-by-doing. You will have to plan, design and implement a web application using two different technologies – PHP and ASP .Net. This project has to be completed by **two or three person teams**. (**Note: Permission to work individually will be granted on a**

case-by-case basis).

Project Minimum Requirements

You can choose to design and implement any web application as long as it has the following key components:

- A back-end database (with a minimum of five tables).
- Dynamic and interactive web pages that enable a user a) to search one or more tables with a minimum of two search criteria, b) add one or more records to one or more tables, c) update/modify one or more records in one or more tables, and d) delete one or more records in one or more tables.
- Secured pages that can be accessed only by authenticated users
- Login functionality (i.e., to authenticate a user)
- Logout functionality (i.e., to terminate an authenticated user's session)
- State Management (i.e., the use of session variables and/or cookies to maintain state)

Project Scope

Keep in mind that you have a limited amount of time to complete the implementations (about 6 weeks for each). Therefore, plan the scope of your project wisely. Your project does not have to include all the functionalities of a commercial web application. To keep your project manageable and doable, limit your application to about 6 functionalities, 8 pages, and 10 tables.

Project Proposal

A project proposal (limit 5 pages) that provides the following information should be submitted on or before **January 29**:

- **Description** – an overall description of the application that outlines its purpose, scope and target users.
- **Functionality** – an itemized list (accompanied by a brief description) of the different functionalities that the application will offer (e.g., for a hotel reservation application, some of the functionalities would be a) search for availability by date and room preference, b) view room options and rates, c) make a reservation, d) receive confirmation, e) modify a reservation, and f) cancel a reservation).
- **Business Rules** – a preliminary list of the rules that will govern your application (e.g., for a student registration system, some applicable business rules would be a student cannot register for a) the same class twice, b) more than 18 credits, and c) classes with overlapping times).
- **Assumptions** – assumptions (if any) that you have made about who will use your application and how they will use it.
- **Web Pages** – a preliminary list of web pages (e.g., welcome/home page, login page, search page, results page, and confirmation page). For each page, a brief description of its purpose and a written or graphic representation of its layout and contents. The proposed flow among the pages should also be included.
- **Validation** – a preliminary list of the validation checks that will be performed on input data.
- **Database Schema** – a preliminary list of normalized tables that will be required to support your application. For each table, its fields (name, data type and description) should be provided. Primary and foreign keys should also be clearly identified.
- **Model Sites**: A list of two to four existing sites on the Web that you plan to use as models for your application.
- **Pseudo use case diagram** - your application's functionalities should be captured in a use case diagram (a sample diagram is posted within Modules → Course Information on Canvas)
- **ERD Diagram**: your database schema should be captured in a high-level conceptual ERD diagram (a sample diagram is posted within Modules → Course Information on Canvas).

Your proposal will be graded on its presentation - organization, format, spelling, grammar – (10%) and content – detail, completeness, clarity – (90%).

Project Implementations

After your proposal is approved, you will implement it twice; first using PHP (due on or before March 14) and then with ASP.Net (due on or before May 9). An updated proposal that includes any changes recommended by me/your IC or made by you (**with my/your IC's approval**) during development will also be due on March 8. **Note: Your implementations should reflect the database schema and functionalities that were approved by me/your IC. We will not accept any unapproved deviations/modifications from the proposal.** Instructions on how to submit the projects will be provided separately.

Project Grading Criteria

Your project proposal will be graded for its professional presentation and completeness. The implementations will be judged on the following:

- Functionality – Dynamic and interactive web pages that enable a user a) to search one or more tables with a minimum of two search criteria, b) add one or more records to one or more tables, c) update/modify one or more records in one or more tables, and d) delete one or more records in one or more tables (55%)
- Security and State - Secured pages that can be accessed only by authenticated users; login functionality (i.e., to authenticate a user); logout functionality (i.e., to terminate an authenticated user's session); state management (i.e., the use of session variables and/or cookies to maintain state) (20%).
- Business Rules and Validation – adherence to the proposed business rules (e.g., should not be able to register for more than 18 credit hours); use of appropriate validation checks (e.g., required fields, correct data types, range checks) to ensure an error-free user experience (8%).
- Database – Minimum of five tables that are normalized; primary keys are defined; referential integrity is set; and all five tables are used by the application (7%).
- Design – general site appearance, clear and logical navigation capability, appropriate choice of controls and images, page titles, consistency in the use of fonts, colors, style and layout, spelling and grammar, intuitive and easy to use interface (5%).
- Completeness - to what extent and how well the proposed functionalities are incorporated; to what extent the project has a professional “look and feel” and is ready for deployment (5%).

Research Presentation

The intent of the research presentations is to provide

- an opportunity to explore a topic that is of interest to you related to the design, development, implementation, use, and management of web applications (e.g., broad topics including security, privacy, encryption, payment systems, accessibility, content management, online auctions, file sharing, VoIP, virtual social networks, mobile applications, cloud computing, cryptocurrencies or more specialized topics such as languages, development frameworks, style sheets, W3C, browsers, web services, search engines, web portals, developer communities), and
- a forum to share your team's research with the class.

Each team will submit a topic (to a forum on Canvas) by **January 31**. Check the topics that have already been submitted before posting yours. There should be no duplication of topics. If you are keen on presenting a certain topic, make sure that you submit it as early as possible.

Both campus and distance students will record their presentations and post them on YouTube. Instructions for recording and posting videos are provided within Modules → Course Information on Canvas. We will make the presentation links and slides available to all students. A written research report is not required. Just submit the slides used in your presentation.

The research presentations (15 to 20 minutes) will be graded on content – depth and breadth of research – (80%) and presentation format and style - organization, clarity, poise, use of appropriate examples/demos – (20%).

Exams

Both exams will be cumulative. Questions for the exams will be based on lectures, handouts, sample programs, assignments, in-class exercises, class discussions and presentations. While the typical format for the exams would be programming questions that require you to analyze and/or write code snippets, other formats such as multiple-choice, short-answer and essay answer questions may also be used.

Teams

You have to indicate your team membership (to a forum on Canvas) by **January 24**. You cannot change membership after the project proposal has been submitted and approved. However, you have the option of leaving a team (permanently) and completing the projects and presentation individually.

By default, all members of a team will receive the same grade for their research presentations, project proposal and implementations. However, this may change based on peer evaluations that each team member is **required to complete** for the research presentation and each project deliverable. A sample peer evaluation form is posted on Canvas.

Working as a team can be challenging. There could be personality conflicts, squabbles over work allocations, problems in scheduling meetings, disputes over strategy, and frustration about a member not doing his/her fair share. Therefore, exercise caution in forming your team. Dealing with team disputes and resolving them amicably will be part of the overall learning experience. If you require my/your IC's intervention over an unsolvable conflict, please contact me/your IC as soon as possible.

Assumptions and Expectations

My assumption is that each one of you is self-motivated to learn and perform well in this course. You have to be willing and prepared to supplement your classroom learning with reference books, peer assistance, and online help. The field of web development is vast and in a constant state of flux. I will be more a facilitator and partner in your learning experience, rather than an all-knowing guru. You are strongly encouraged to ask questions, share your knowledge/ideas, make suggestions, express views/opinions, and actively participate in class discussions.

Academic Honesty

All work in this course must be completed in a professional manner and should be consistent with CSU's Policy on Academic Integrity - <https://tilt.colostate.edu/integrity/knowTheCode/>. You are required to complete the assignment, hands-on exercises and exams individually and the research presentation and projects in teams of no more than three. Plagiarism or copying someone else's work and submitting it as your own is a serious violation of University Policy. The penalty for academic misconduct is zero credit for the assignment, exercise, exam or project work. In addition, at my discretion, you could receive letter grade reductions or an F grade for the course (refer to CIS Department - Academic Integrity Policy on Canvas). **Note:** The above penalties will apply to all students involved in a misconduct incident (i.e., those who copied and those who allowed their work to be copied).

Administrative

Course requirements have to be submitted by the due date and time. No exceptions – work submitted after a deadline will not be graded. If you have a critical family and/or medical emergency, I will consider granting an extension provided you contact me in person or by email/phone before the deadline.

Campus students are expected to attend class sessions. It is understandable for you to miss a class or two. But a pattern of regular absences will result in a forfeiture of points. Specifically, you would lose 5% of the points allocated for hands-on-exercises for each unexcused absence.

Check the University Calendar for drop and withdrawal dates. Your absence in class (or, failure to view a class session) is no excuse for ignorance of any announcements including changes to the course schedule. It is your responsibility to find out about materials covered, homework assigned and/or announcements made if you happen to miss a class.

Any student with disabilities or other special needs, who requires special accommodation in this course, is invited to share these concerns or requests with the instructor as soon as possible.

Tentative Schedule

Week	Date	Topic	Due
1	01/22	Introduction to the course; HTML - Basics	
	01/24	CSS - Basics	Group Membership
2	01/29	CSS - Layout	Project Proposal
	01/31	CSS – Graphic Design	Research Presentation Topic
3	02/05	HTML – Tables and Forms	
	02/07	HTML - Multimedia and Java Script	
4	02/12	PHP – Basics: Tags and Functions	
	02/14	PHP – Conditional Processing and Loops	
5	02/19	PHP – Accessing Databases	HTML Assignment
	02/21	PHP – Search	
6	02/26	PHP - Validation	
	02/28	PHP – Insert/Edit/Delete	
7	03/05	PHP – Cookies & Session Variables	
	03/07	PHP – Security	
8	03/12	PHP – Additional Topics	
	03/14	Midterm Exam	PHP Implementation
9	03/19	Spring Break	
	03/21		
10	03/26	ASP – MVC Basics	
	03/28		
11	04/02	ASP –C# Primer	
	04/04		
12	04/09	ASP - Razor	
	04/11		
13	04/16	ASP – Entity Framework	
	04/18		
14	04/23	ASP – Displaying/Searching Data	
	04/25		
15	04/30	ASP – Creating, Editing, Deleting Data	
	05/02		
16	05/07	ASP – Authentication and Authorization	
	05/09		ASP Implementation
17	05/16	Final Exam (9:40 to 11:40 AM)	